Executive Summary

Overview

This project analyzes service activity within the Emergency Food Network (EFN) partner system using annual data from 2014 to 2024. The analysis focuses on how service reach, repeat utilization, and food distribution patterns changed over time and across service units.

The primary goal is descriptive: to characterize the structure and evolution of recorded service activity within the observed EFN network. The project does not attempt to estimate causal effects, evaluate program performance, or measure underlying food insecurity in the broader community.

In addition to the descriptive analysis, a limited time-series extension is included to assess whether the aggregate annual service series supports a cautious baseline projection. Given the short annual sample and the instability observed after 2020, this extension is treated as secondary to the broader operational analysis.

Key Findings from the Descriptive Analysis

Several findings emerge from the historical data:

- Total service activity increased substantially over the sample period, especially after 2020.
- Growth in total service encounters outpaced growth in unduplicated clients, indicating increased repeat utilization over time.
- Average activity per service unit also rose, suggesting that growth was not driven solely by changes in network coverage.
- Age composition was relatively stable across most of the sample, with some temporary disruption around 2020.
- Service activity and distribution patterns varied meaningfully across service units, indicating substantial heterogeneity in scale and utilization across the network.

Taken together, these patterns suggest that EFN partner activity became both larger in scale and more intensive over time, particularly in the post-2020 period.

Time-Series Extension

To complement the descriptive analysis, the project constructs an aggregate annual series of total service encounters and evaluates whether a simple univariate forecasting exercise is methodologically defensible.

The annual series is relatively stable before 2020, then becomes much more volatile. A sharp increase in 2020 is followed by a decline in 2021 and strong growth thereafter. Because this short annual series does not exhibit a stable, repeating dynamic pattern, the selected baseline model produces flat point projections over the forecast horizon and wide prediction intervals.

These results should not be interpreted as evidence that future service demand will remain unchanged. Rather, they indicate that the historical data do not support projecting sustained increases or decreases with confidence. The time-series component is therefore best interpreted as a conservative planning baseline under substantial uncertainty.

Key Limitations

Several limitations are important:

- The data describe recorded service activity within observed service units, not underlying food insecurity in the full community.
- The annual time series is short, limiting statistical power and the complexity of defensible forecasting methods.
- The post-2020 period is structurally unstable, reducing the reliability of trend extrapolation.
- The time-series extension does not incorporate external drivers such as policy changes, funding conditions, or macroeconomic shifts.

Bottom Line

This project is best understood as an applied service-demand and utilization analysis with a limited time-series extension. Its main contribution is a careful descriptive account of how service activity changed over time and across the EFN network, supported by transparent discussion of data structure, measurement limits, and uncertainty.

The baseline projection is included as a cautious extension, not as the centerpiece of the analysis.